MINDO - Project Documentation Guidelines (Phase 1 & Phase 3)

Brand Name: MINDO (Snack Smart. Live Sharp.)
Project Title: Digital Marketing & Functional Snack Web E-Commerce Platform
Target Audience: Youth Aged 18–30 (Students, Early-Career Professionals, Active Individuals)
Brand Tagline: "Fuel Your Next Move."

Project Team & Roles

- **Mohamed Awad** — Team Leader / Project Manager & Media Buyer Lead
- **Rawan Tarek** — Requirements Analyst & Content Strategist
- **Menna Khaled** — System & UI/UX Designer
- **Jasmine Alaa** — Full Stack & Database Developer

Part 1: Phase 1 - Project Planning & Management

Phase Submission Deadline: 16 Oct 2026

1.1 Project Proposal

Overview

MINDO is a modern Egyptian functional-snack brand and integrated digital e-commerce ecosystem designed specifically for dynamic individuals aged 18–30. By combining nutritional utility (focus, sustained energy, high protein) with an effortless, mobile-first purchasing experience, MINDO fulfills the demand for healthy, quick-access fuel in a fast-paced environment. The core platform functions as both a customer-facing shop and an internal campaign/analytics manager under the brand promise *"Fuel Your Next Move."*

Core Objectives

1. **Digital Brand Establishment:** Launch an intuitive, visually distinct direct-to-consumer (D2C) web storefront that builds strong brand equity.
2. **Sales Conversion Optimization:** Streamline the checkout workflow to achieve a target conversion rate of $\ge 3.5\%$ from incoming social and paid ad traffic.
3. **Data-Driven Campaign Tracking:** Measure real-time customer acquisition costs (CAC), ad source conversions, and inventory turnover via an integrated admin dashboard.
4. **Customer Retention:** Deliver personalized snack suggestions categorized by daily functional needs (e.g., Focus, Energy, Recovery).

Scope Management

In-Scope

- Responsive web application (Customer Storefront + Admin Dashboard).
- Product catalog filtering, cart management, and multi-step checkout workflow.
- Relational database architecture for users, orders, inventory, and marketing campaigns.
- Digital marketing campaign integration (tracking URL parameters and conversion funnels).
- Role-based access control (Customers vs. System Administrators).

Out-of-Scope (Phase 1–3)

- Native mobile applications (iOS/Android) — deferred to post-launch roadmap.
- Automated physical warehouse robotics integration.
- Dynamic multi-currency pricing (v1.0 restricted to EGP).

1.2 Project Plan & Timeline

The project is structured across five sequential phases running from October through December 2026.

Phase	Description	Key Deliverables	Start Date	Deadline
Phase 1	Project Planning & Management	Project Proposal, Timeline, Task Assignment, Risk Matrix, KPIs	02 Oct 2026	16 Oct 2026
Phase 2	Requirements Gathering & Analysis	Stakeholder Analysis, User Stories, Functional & Non-Functional Specs	09 Oct 2026	16 Oct 2026
Phase 3	System Analysis & Design	Architecture Diagrams, Database Schema/ERD, DFDs, Sequence Diagrams, Wireframes	09 Oct 2026	06 Nov 2026
Phase 4	Implementation & Execution	Source Code, Backend APIs, Database Integration, Version Control (GitHub)	16 Oct 2026	30 Nov 2026
Phase 5	Testing, QA & Final Delivery	Test Execution Suite, User Manual, Technical Docs, Final Pitch Presentation	16 Oct 2026	04 Dec 2026

1.3 Task Assignment & Team Responsibilities

To ensure clear accountability, team roles map directly to specialized workstreams across the development lifecycle:

Mohamed Awad — Team Leader / Project Manager & Media Buyer Lead

- Leads overall sprint planning, task delegation, and cross-functional synchronization.
- Oversees digital marketing strategies, media buying budget allocation, and campaign tracking models.
- Conducts continuous risk monitoring and alignment with DEPI program milestones.

Rawan Tarek — Requirements Analyst & Content Strategist

- Gathers functional and business requirements across user personas.
- Conducts market research and stakeholder alignment for the target demographic (18–30 age bracket).
- Crafts campaign content mapping, brand voice copy, and product catalog descriptions.

Menna Khaled — System & UI/UX Designer

- Designs structural system models, including DFDs, Sequence Diagrams, and Use Case Diagrams.
- Leads visual design and responsive UI wireframing in accordance with MINDO visual guidelines.
- Ensures accessibility, consistent color applications (Deep Navy, Lime Green, Orange Accent), and seamless UX design.

Jasmine Alaa — Full Stack & Database Developer

- Designs normalized relational database models (ERD, Physical Schema).
- Develops core backend RESTful APIs and integrates storefront business logic.
- Establishes GitHub workflow, CI/CD pipelines, and cloud hosting infrastructure.

1.4 Risk Assessment & Mitigation Plan

Risk Description	Probability	Impact	Risk Level	Mitigation Strategy	Owner
Scope Creep / Feature Creep	Medium	High	High	Enforce strict MVP boundary. Non-essential user features logged in post-launch backlog.	Mohamed Awad
Tight Phase Deadlines	High	Medium	High	Conduct bi-weekly team syncs. Maintain 3-day buffer ahead of 16 Oct, 6 Nov, and 30 Nov milestones.	Mohamed Awad

Risk Description	Probability	Impact	Risk Level	Mitigation Strategy	Owner
Database Bottlenecks under Peak Traffic	Low	High	Medium	Implement proper field indexing, query optimization, and dynamic connection pooling.	Jasmine Alaa
Brand Identity Inconsistency	Medium	Medium	Medium	Maintain locked UI Kit and Design System token file matching brand guidelines.	Menna Khaled
Integrations & Deployment Errors	Low	Medium	Low	Maintain strict GitHub Pull Request code reviews before merging into `main` branch.	Jasmine Alaa

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1.5 Key Performance Indicators (KPIs)

The success of the MINDO digital platform is evaluated against three standard performance pillars:

System Performance & Stability

- **Uptime:** $\ge 99.5%$ target platform availability.
- **Page Load Latency:** < 2.0 seconds average response time across standard 4G mobile connections.
- **API Response Time:** $< 200\text{ ms}$ for core catalog query endpoints.

User Engagement & Conversion Metrics

- **Checkout Conversion Rate:** $\ge 3.5%$ conversion rate from active site sessions.
- **Click-Through-Rate (CTR):** $\ge 4.0%$ across active ad campaigns targeting storefront product pages.
- **Cart Abandonment Rate:** Maintained below $60%$.

Code Quality & Execution

- **Version Control Discipline:** $100%$ code contributions executed via reviewed GitHub Pull Requests.
- **Test Coverage:** $\ge 80%$ functional test suite pass rate prior to Phase 5 release.

Part 2: Phase 3 - System Analysis & Design

Phase Submission Deadline: 06 Nov 2026

2.1 Problem Statement & Objectives

Problem Statement

Young Egyptian professionals and students lack quick, accessible, and high-quality functional snacking options tailored to their daily lifestyles. Existing market options are either traditional high-sugar confectioneries that cause energy crashes or expensive imported health products with limited online availability. Furthermore, current local channels lack an integrated, modern e-commerce experience tailored to mobile-first digital natives.

Proposed System Solution

The MINDO Web E-Commerce Platform bridges this gap by providing an online store optimized for high-speed browsing and targeted functional discovery. The platform allows users to browse products aligned with specific functional needs (Study Focus, Sustained Energy, Fitness Recovery), place instant orders, track delivery, and engage with seasonal marketing promotions.

2.2 System Requirements Specification

Functional Requirements (FR)

FR-1: User Management & Authentication

- **FR-1.1:** Account creation via email/password validation.
- **FR-1.2:** Secure user authentication supporting encrypted sessions and persistent tokens.
- **FR-1.3:** Customer profile management (address book, order history, preferred snack categories).
- **FR-1.4:** Role-Based Access Control distinguishing Customers from System Administrators.

FR-2: Product Catalog & Functional Discovery

- **FR-2.1:** Interactive product catalog display with clear pricing and stock badges.
- **FR-2.2:** Filtering options by functional target (*Focus/Study*, *Energy*, *High Protein*).
- **FR-2.3:** Detailed product pages detailing ingredient profiles and nutritional facts.

FR-3: Cart & Checkout Engine

- **FR-3.1:** Dynamic guest and authenticated shopping cart manipulation (add, update quantity, remove).
- **FR-3.2:** Automated order summary calculations (subtotal, shipping fees, tax).
- **FR-3.3:** Seamless checkout workflow supporting standard online transactions and Cash-on-Delivery (COD).

FR-4: Admin Operations & Marketing Analytics

- **FR-4.1:** Inventory management module to add, edit, or adjust stock levels.
- **FR-4.2:** Campaign tracking module to log traffic sources, coupon usage, and conversion metrics.
- **FR-4.3:** Order status tracking dashboard (Pending, Processing, Shipped, Delivered).

Non-Functional Requirements (NFR)

NFR-1: Security & Compliance

- All traffic enforced over HTTPS with SSL/TLS encryption.

- Sensitive passwords stored using strong salted `bcrypt` hashing algorithms.
- Protected API endpoints defense against SQL Injection (SQLi) and Cross-Site Scripting (XSS).

NFR-2: Usability & Mobile Accessibility

- Mobile-first responsive UI grid system compatible across modern viewports (320px to 1920px).
- Design System adherence to MINDO visual standards (Deep Navy, Lime Green, Orange Accent).

NFR-3: Reliability & Performance

- Platform performance optimized for sub-2.0 second page load time.
- Scalable relational database architecture capable of supporting concurrent customer sessions.

2.3 Software Architecture & Technology Stack

The platform utilizes a modern 3-Tier Layered Architectural Pattern organized cleanly across Presentation, Application Logic, and Persistence layers.+----------+

| PRESENTATION LAYER |

| React.js Web Client / HTML5 / Tailwind CSS (MINDO Brand Theme) |

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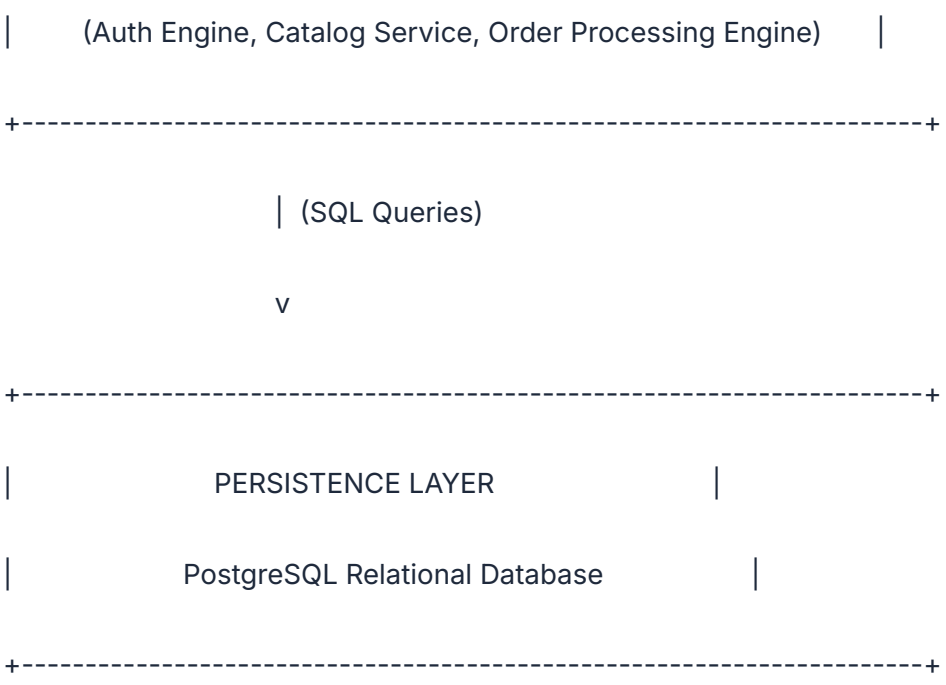
| (JSON / REST API)

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| APPLICATION LAYER |

| Node.js & Express.js REST APIs / Business Logic |



Technology Stack Summary

Layer	Selected Technology	Technical Justification
Frontend UI	React.js / Tailwind CSS	Component-driven framework for rapid rendering and state management; easily customized with MINDO brand design tokens.
Backend API	Node.js (Express.js)	Non-blocking, event-driven architecture well-suited for high-concurrency e-commerce operations.
Database	PostgreSQL	Robust transactional integrity (ACID compliance) required for precise order processing and stock tracking.

Layer	Selected Technology	Technical Justification
Version Control	GitHub	Collaboration, code reviews, and structured repository branch management.
Deployment	Vercel (Frontend) / Render (Backend)	Cost-effective, high-reliability deployment pipelines with automated SSL provisioning.

2.4 Database Design & Data Modeling

Entity-Relationship Diagram (ERD) Overview

The data architecture relies on a normalized relational model (3NF) designed by Jasmine Alaa.

Relational Schema Definition

1. Users Table (`users`)

- `user_id` (PK, INT, AUTO_INCREMENT)
- `full_name` (VARCHAR 100, NOT NULL)
- `email` (VARCHAR 150, UNIQUE, NOT NULL)
- `password_hash` (VARCHAR 255, NOT NULL)
- `phone_number` (VARCHAR 20)
- `role` (ENUM('customer', 'admin'), DEFAULT 'customer')
- `created_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

2. Products Table (`products`)

- `product_id` (PK, INT, AUTO_INCREMENT)
- `title` (VARCHAR 150, NOT NULL)
- `description` (TEXT)
- `category` (ENUM('Focus', 'Energy', 'Protein'), NOT NULL)
- `price` (DECIMAL(10,2), NOT NULL)
- `stock_quantity` (INT, DEFAULT 0)

- `nutritional_info` (JSON)
- `created_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

3. Orders Table (`orders`)

- `order_id` (PK, INT, AUTO_INCREMENT)
- `user_id` (FK → `users.user_id`, NOT NULL)
- `total_amount` (DECIMAL(10,2), NOT NULL)
- `order_status` (ENUM('Pending', 'Processing', 'Shipped', 'Delivered', 'Cancelled'), DEFAULT 'Pending')
- `shipping_address` (TEXT, NOT NULL)
- `created_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

4. Order Items Table (`order_items`)

- `order_item_id` (PK, INT, AUTO_INCREMENT)
- `order_id` (FK → `orders.order_id` ON DELETE CASCADE, NOT NULL)
- `product_id` (FK → `products.product_id`, NOT NULL)
- `quantity` (INT, NOT NULL)
- `unit_price` (DECIMAL(10,2), NOT NULL)

5. Campaigns Table (`campaigns`)

- `campaign_id` (PK, INT, AUTO_INCREMENT)
- `campaign_name` (VARCHAR 100, NOT NULL)
- `source_channel` (VARCHAR 50, NOT NULL)
- `promo_code` (VARCHAR 30, UNIQUE)
- `conversion_count` (INT, DEFAULT 0)
- `created_at` (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

2.5 System Behavior & Diagrammatic Modeling

Use Case Diagram Description

The platform defines two primary system actors:

1. **Customer:** Registers account, browses functional catalog, filters snacks, manages shopping cart, executes checkout, and tracks order fulfillment.
2. **Admin:** Authenticates to administrative dashboard, manages product listings/inventory, reviews customer order queues, and monitors marketing campaign conversion stats.

Data Flow Diagram (DFD)

Level 0 (Context Diagram)

The central MINDO E-Commerce System acts as the core interface between external entities: **Customer**, **Admin**, and **Payment System**.

- **Customer** provides Registration Details, Search Queries, and Order Inputs; receives Catalog Views, Cart Summaries, and Order Confirmations.
- **Admin** provides Inventory Updates and Campaign Parameters; receives System Metrics, Analytics, and Order Reports.

Level 1 (Process Breakdown)

1. **1.0 User Authentication Process:** Validates credentials against `users` table; issues access tokens.
2. **2.0 Catalog Management Process:** Reads product data from `products` table based on search filters.
3. **3.0 Order Engine Process:** Calculates totals, creates record in `orders` and `order_items`, and updates stock levels in `products`.
4. **4.0 Marketing & Analytics Process:** Correlates traffic parameters with sales to update `campaigns` performance table.

Sequence Diagram: Purchase & Checkout Execution

1. **Customer** adds items to cart and submits checkout request via Frontend Client.
2. **Frontend Client** posts payload to Backend API (`POST /api/orders`).
3. **Backend API** queries Database to verify product inventory levels (`products` table).
4. **Database** confirms stock availability.
5. **Backend API** locks inventory, generates order records in `orders` / `order_items`, and calculates final balance.
6. **Backend API** returns `201 Created` status with Order Confirmation details to Frontend Client.

7. **Frontend Client** displays success notification and order tracking number to Customer.

Class Diagram Description

Core class representations within backend architecture:

- `User`: Base class handling credentials and personal data.
- `Customer` *(extends User)*: Methods: `browseCatalog()`, `addToCart()`, `checkout()`.
- `Admin` *(extends User)*: Methods: `updateStock()`, `createCampaign()`, `viewAnalytics()`.
- `Product`: Attributes: `productId`, `title`, `price`, `stockQuantity`; Methods: `checkAvailability()`, `updatePrice()`.
- `Order`: Composition of `OrderItem` objects; Methods: `calculateTotal()`, `updateStatus()`.
- `Campaign`: Attributes: `campaignId`, `sourceChannel`, `conversionCount`; Methods: `trackConversion()`.

2.6 UI/UX Design Guidelines & Brand System

The user interface design, developed by Menna Khaled, reflects MINDO's energetic visual identity.

Palette Standards

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| PRIMARY: Deep Navy (#0A192F) |

| Usage: Core backgrounds, navigation headers, primary structural text |

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| ACCENT: Lime Green (#32CD32) |

| Usage: Calls to Action (CTA), purchase buttons, success badges |

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| HIGHLIGHT: Orange Accent (#FF8C00) |

| Usage: Flash sales, energy category badges, discount alerts |

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Color Palette Matrix

Color Token	Hex Code	Visual Identity Role	Usage Context
Deep Navy	`#0A192F`	Foundation & Luxury	Primary background, structural headers, high-contrast dark elements.
Lime Green	`#32CD32`	Energy & Freshness	Primary CTA buttons ("Add to Cart", "Checkout Now"), availability badges.
Orange Accent	`#FF8C00`	Warmth & Attention	Promotional banners, discount highlights, functional tags (*Energy Kick*).
Light Slate	`#F8FAFC`	Surface Neutral	Card background containers, section dividers, input fields.

Typography & Structure Rules

- **Font Family:** Clean sans-serif (`Inter`) providing high legibility on small screens.
- **Layout Grid:** Responsive 12-column desktop grid collapsing down to a single-column layout on mobile viewports.
- **Card Design:** Snack products displayed in clean card layouts emphasizing functional benefit badges (*Focus*, *Energy*, *Protein*), price, and instant "Add to Cart" CTAs.